

EXP NO: 8

TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Signup & Login
- Viewing and Managing Playlists
- Fetching Real-time Metadata
- Editing playlists (rename, reorder, record)
- Creating smart audio playlists based on categories (mood, genre, artist, etc.)

2. Define User Interactions

- Each test case simulates a real user behaviour (e.g., logging in, renaming a playlist, adding a song).

3. Design Happy Path Test Cases

- Focused on validating that all features function as expected under normal conditions.
- Example: User logs in successfully, adds item to playlist, or creates a category-based playlist.

4. Design Error Path Test Cases

- Simulate negative or unexpected scenarios to test robustness and error handling.
- Example: Login fails with invalid credentials, save fails when offline, no recommendations found.

5. Break Down Steps and Expected Results

- Each test case contains step-by-step actions and a corresponding expected outcome.
- Ensures clarity for both testers and automation scripts.

6. Use Clear Naming and IDs

- Test cases are named clearly (e.g., TC01 – Successful Login, TC10 – Save Playlist Fails).
- Helps in quick identification and linking to user stories or features.

7. Separate Test Suites

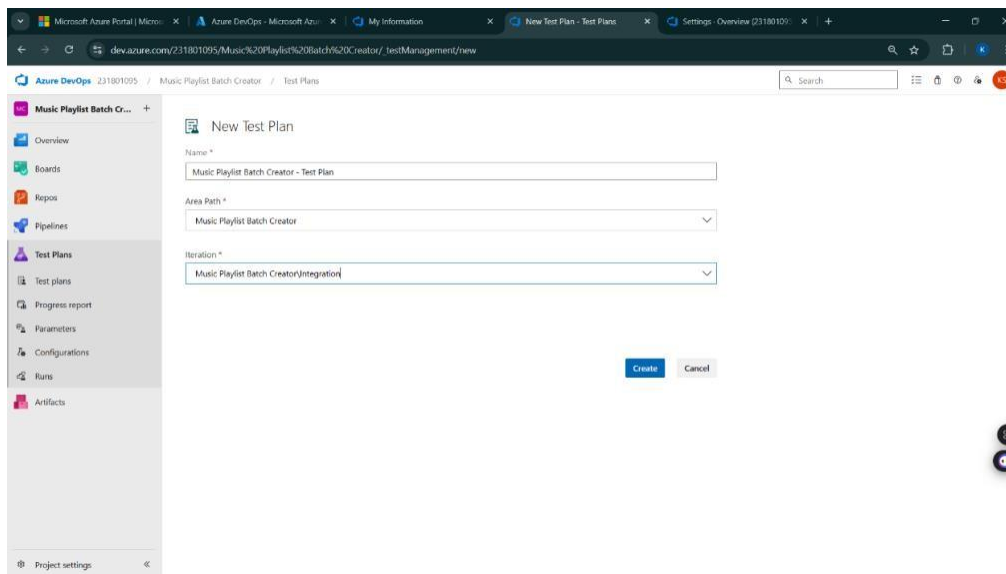
- Grouped test cases based on functionality (e.g., Login, Playlist Editing, Recommendation System).

- Improves organization and test execution flow in Azure DevOps.

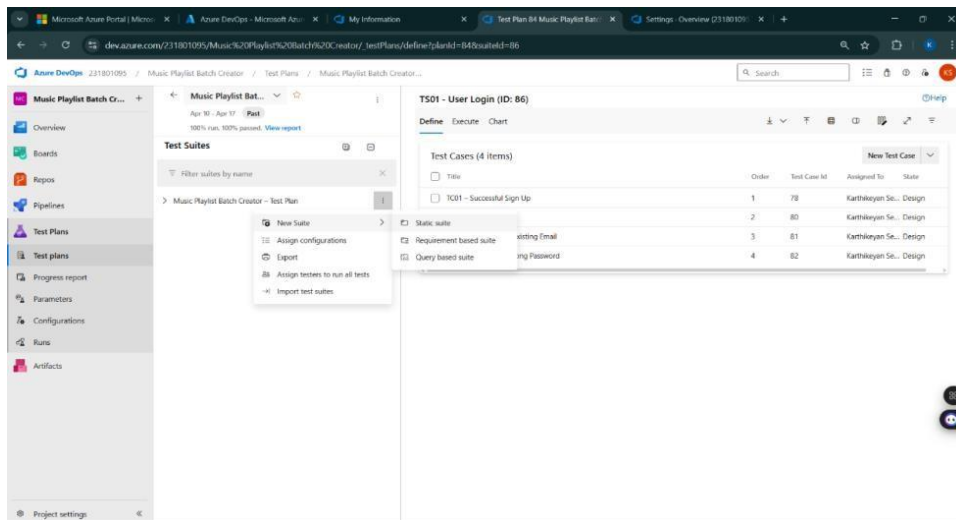
8. Prioritize and Review

- Critical user actions are marked high-priority.
- Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Complaint management system – Test Plans

USER STORIES

- As a user, I want to sign up and log in securely so that I can access my playlists (ID: 79).
- As a user, I need to see my playlist in one place (ID: 76).
- As a user, I should be able to create an audio playlist as needed (ID: 73).
- As a user, I should be able to rename, record, and change the playlist (ID: 68).
- As a user, I need to have real-time metadata (ID: 65).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. TC01 – Successful Sign Up

- **Action:**
 - Go to the Sign-Up page.
 - Enter valid name, email, and password.
 - Click "Sign Up".
- **Expected Results:**
 - Sign-Up form is displayed.
 - Fields accept values without error.
 - Account is created, and the user is redirected to the dashboard.
- **Type:** Happy Path

2. TC02 – Secure Login

- **Action:**
 - Go to the Login page.
 - Enter valid email and password.
 - Click on "Login".
- **Expected Results:**
 - Login form is displayed.
 - Fields accept data without error.
 - User is logged in and redirected to the dashboard.
- **Type:** Happy Path

3. TC03 – Sign Up with Existing Email

- **Action:**
 - Go to the Sign-Up page.
 - Enter a name and an already registered email.
 - Click on "Sign Up".
- **Expected Results:**

- Fields accept data.
- Error message "Email already registered" is displayed.
- **Type:** Error Path

4. TC04 – Login with Wrong Password

- **Action:**
 - Go to the Login page.
 - Enter valid email and incorrect password.
 - Click on "Login".
- **Expected Results:**
 - Input is accepted.
 - Error message "Invalid username or password" is shown.
- **Type:** Error Path

Test Suite: TS02 - Complaint Submission & Viewing (ID: 87)

1. TC05 – Submit Complaint Successfully

- **Action:**
 - Login successfully.
 - Navigate to "Submit Complaint".
 - Enter complaint details and category.
 - Click "Submit".
- **Expected Results:**
 - Complaint form is displayed.
 - Complaint is submitted successfully.
 - Confirmation message is shown.
- **Type:** Happy Path

2. TC06 – Complaint Submission Failure

- **Action:**
 - Disconnect internet connection.
 - Navigate to "Submit Complaint".
 - Fill in complaint details.
 - Click "Submit".
- **Expected Results:**
 - Network connection is unavailable.
 - Error message "Unable to submit complaint" is displayed.
- **Type:** Error Path

Test Suite: TS03 - Complaint Status Tracking (ID: 88)

1. TC07 – View Complaint Status

- **Action:**
 - Login to the system.
 - Navigate to "My Complaints".
 - Select a submitted complaint.
- **Expected Results:**
 - Complaint details and current status are displayed.
 - Status updates are shown in real time.
- **Type:** Happy Path

2. TC08 – Complaint Status Not Updating

- **Action:**
 - Open complaint status page.

- Refresh or check for updates.
- **Expected Results:**
 - Status remains unchanged or shows fallback message.
 - Error message "Unable to fetch latest status" may appear.
- **Type:** Error Path

Test Suite: TS04 - Complaint Management (ID: 89)

1. TC09 – Edit Complaint Successfully

- **Action:**
 - Navigate to "My Complaints".
 - Select a complaint.
 - Click "Edit".
 - Update complaint description.
 - Click "Save".
- **Expected Results:**
 - Complaint details are updated successfully.
- **Type:** Happy Path

2. TC10 – Edit Complaint with Blank Description

- **Action:**
 - Select a complaint.
 - Remove complaint description.
 - Click "Save".
- **Expected Results:**
 - Error message "Complaint description cannot be empty" is displayed.
- **Type:** Error Path

3. TC11 – Update Complaint Priority

- **Action:**
 - Open complaint details.
 - Change complaint priority (Low/Medium/High).
 - Click "Update".
- **Expected Results:**
 - Complaint priority is updated and saved successfully.
- **Type:** Happy Path

4. TC12 – Complaint Update Failure

- **Action:**
 - Login and open complaint details.
 - Disconnect internet or simulate server failure.
 - Update complaint information.
 - Click "Save".
- **Expected Results:**
 - Error message "Failed to update complaint. Please check your connection" is shown.
- **Type:** Error Path

Test Suite: TS05 - Complaint Resolution & Feedback (ID: 90)

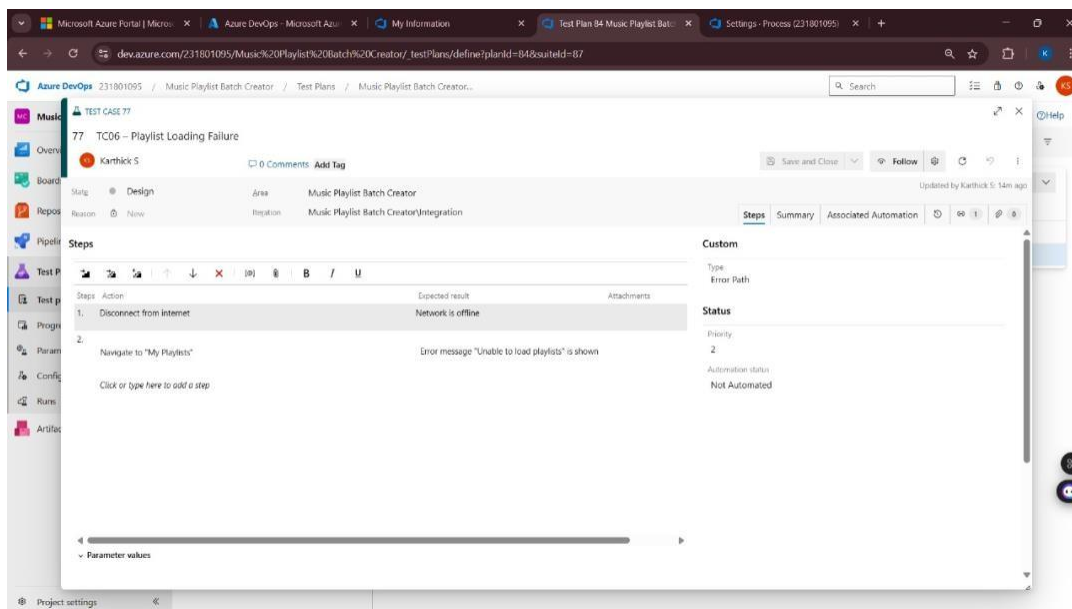
1. TC13 – Submit Feedback for Resolved Complaint

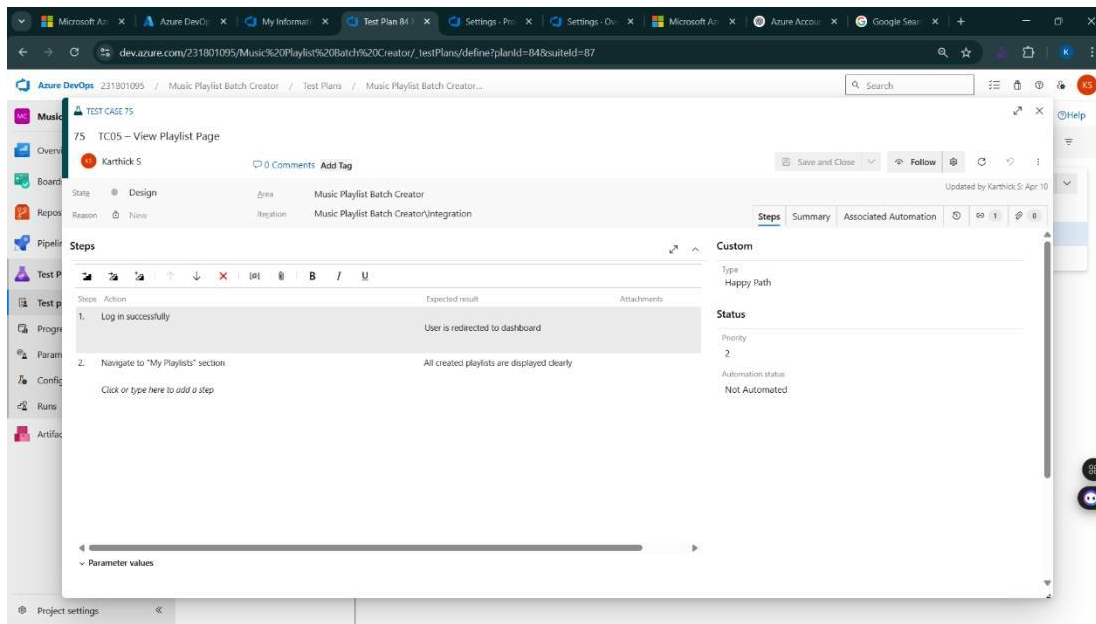
- **Action:**
 - Login with valid credentials.
 - Navigate to resolved complaints.
 - Select a resolved complaint.
 - Enter feedback and rating.
 - Click "Submit Feedback".
- **Expected Results:**
 - Feedback is submitted successfully.
 - Confirmation message is displayed.
- **Type:** Happy Path

2. TC14 – Feedback Submission Failure

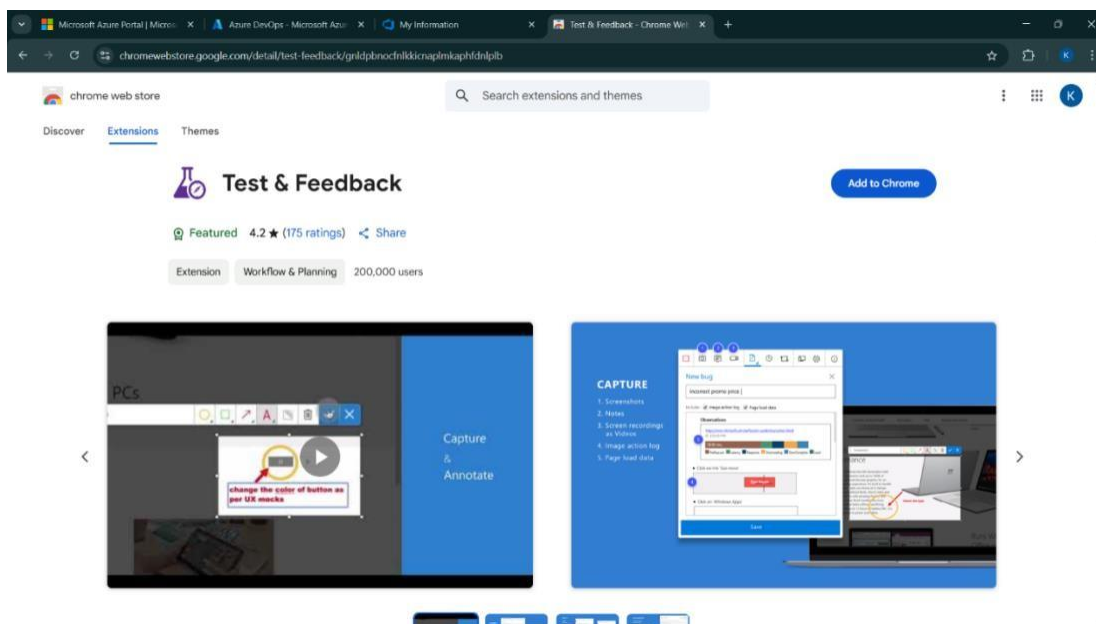
- **Action:**
 - Login successfully.
 - Open feedback section.
 - Leave feedback field blank or enter invalid input.
 - Click "Submit Feedback".
- **Expected Results:**
 - Error message "Feedback cannot be empty" or "Invalid input provided" is displayed.
- **Type:** Error Path

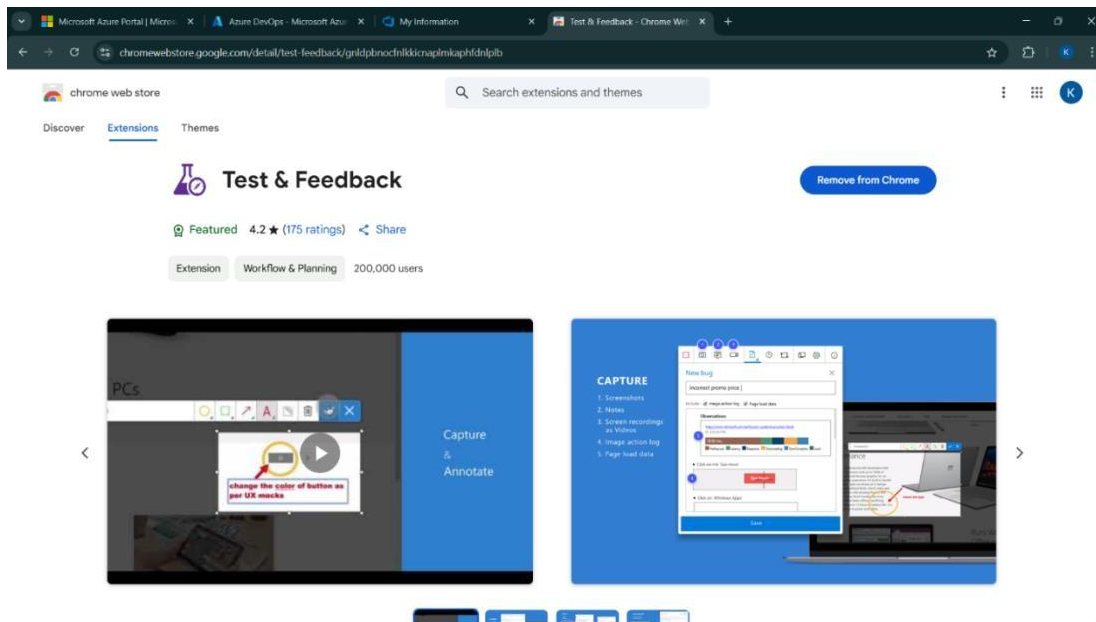
Test Cases



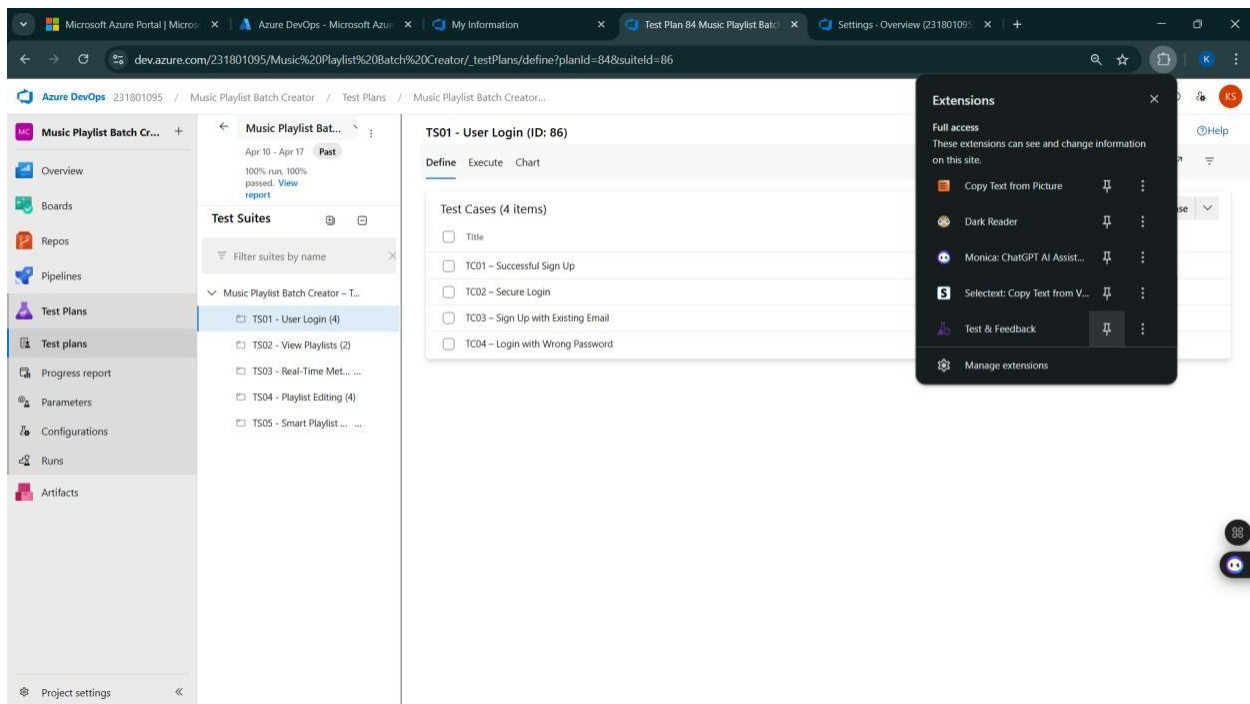


4. Installation of test





Test and feedback
Showing it as an extension



5. Running the test cases

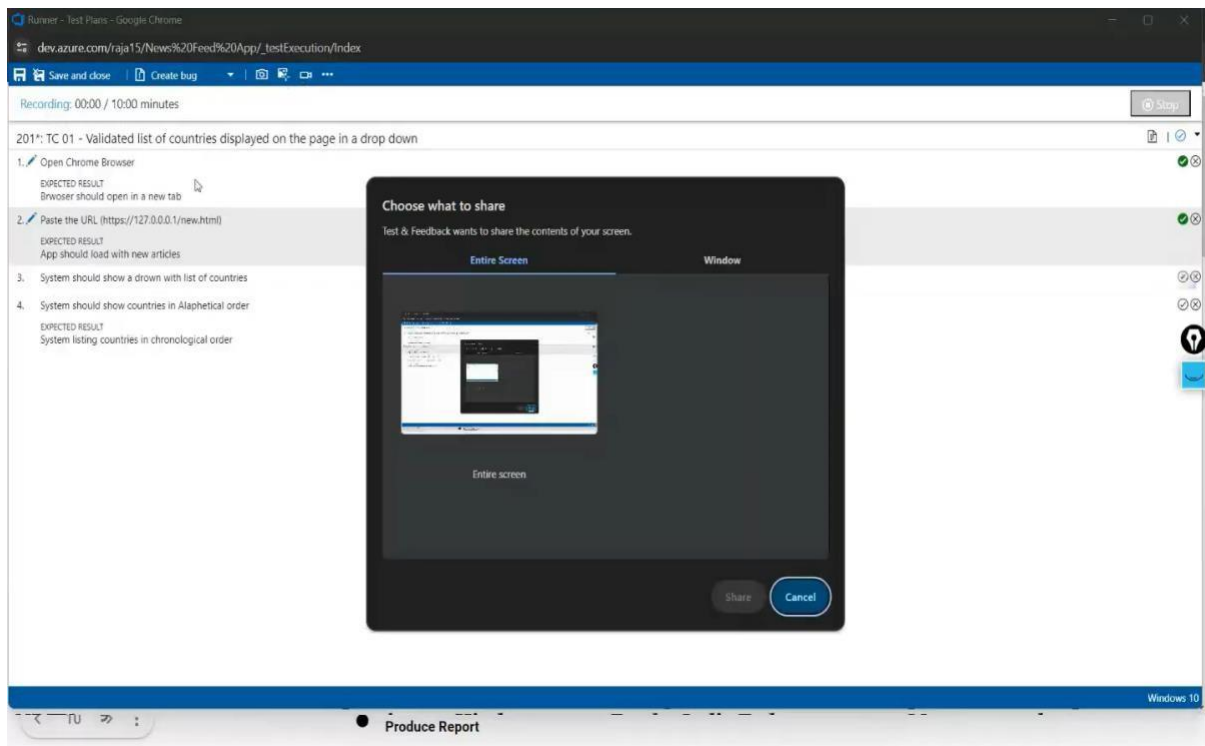
The screenshot shows the Azure DevOps Test Plans interface. The left sidebar contains navigation options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays the 'Music Playlist Batch Creator' test plan with a 'Past' status and a 'View report' link. Below this, the 'Test Suites' section shows a list of suites: TS01 - User Login (4), TS02 - View Playlists (2), TS03 - Real-Time Met..., TS04 - Playlist Editing (4), and TS05 - Smart Playlist... The 'TS02 - View Playlists (2)' suite is selected, showing a list of test points: TC05 - View Playlist Page and TC06 - Playlist Loading Failure. A context menu is open for TC05, showing options: View execution history, Mark Outcome, Run, Reset test to active, Edit test case, Assign tester, and View test result. The 'Run' option is selected, showing a sub-menu with 'Run for web application', 'Run for desktop application', and 'Run with options'. The 'Run for web application' option is selected.

Title	Outcome	Order	Test Case Id	Configuration	Tester
TC05 - View Playlist Page	Passed	1	75	Windows 10	Mallu karthick B...
TC06 - Playlist Loading Failure	Passed	2	77	Windows 10	Mallu karthick B...

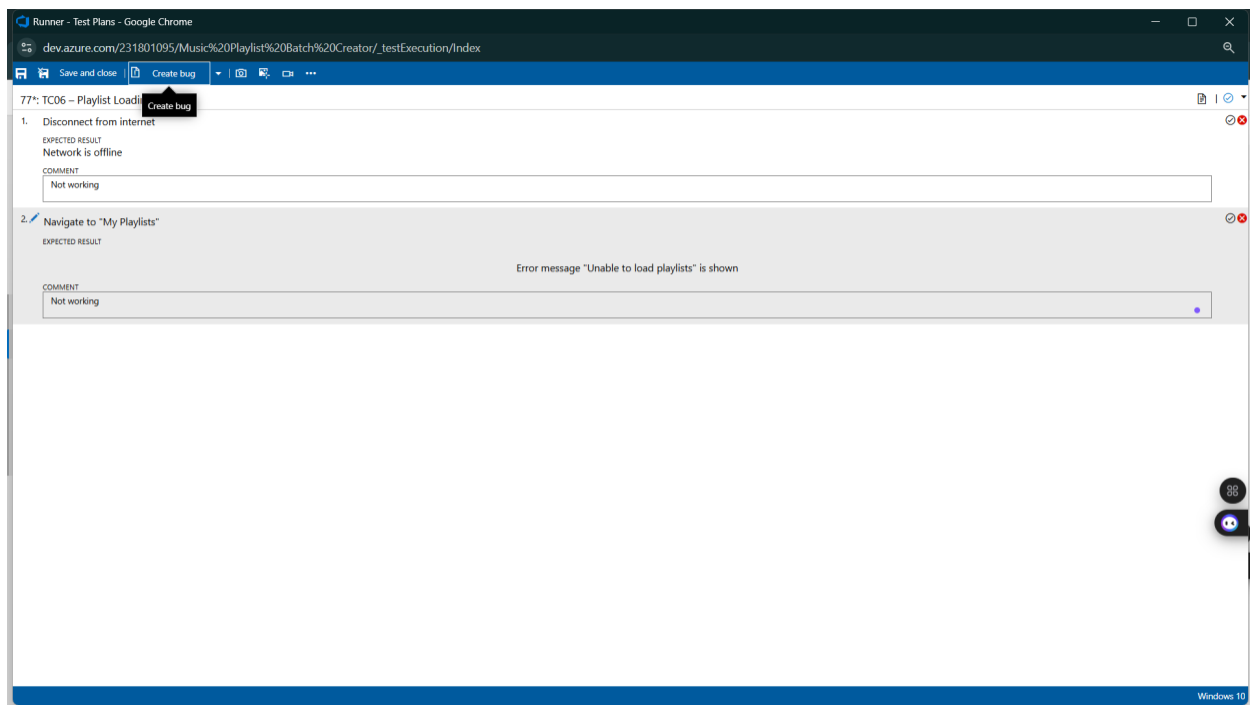
The screenshot shows the 'Runner - Test Plans - Google Chrome' window displaying the test execution results for '75: TC05 - View Playlist Page'. The results are as follows:

- 1. Log in successfully
EXPECTED RESULT
User is redirected to dashboard
- 2. Navigate to "My Playlists" section
EXPECTED RESULT
All created playlists are displayed clearly

6. Recording the test case



7. Creating the bug



Runner - Test Plans - Google Chrome

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testExecution/Index

77% TC06 - Playlist Loading Failure

1. Disconnect from internet

NEW BUG

TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Unassigned 0 comments Add tag TB01 - Playlist loading spinner keeps spinning indefinitely on poor network Save & Close

2. Navigate

State: New Reason: New Area: Music Playlist Batch Creator Iteration: Music Playlist Batch Creator

Details

Repro Steps

18-04-2025 03:23 Bug filed on "TC06 - Playlist Loading Failure"

Step no. Result Title

1. Failed Disconnect from internet

Expected Result

Network is offline

Comments: Page Not loading

2. Failed Navigate to "My Playlists"

Expected Result

Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

Planning

Resolved Reason

Story Points

Priority

2

Severity

3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

+ Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

+ Add link

Add an existing work item as a parent

Tested By

77 TC06 - Playlist Loading Failure Updated 10-04-2025. Design

System Info

Found in Build

Microsoft Azure Portal

Azure DevOps - Music Playlist Batch Creator

Test Plans - Runs

Run 48 - TS02 - View Playlists (Manual) / TC06 - Playlist Loading Failure

92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Unassigned 0 comments Add tag Save & Close Follow

Updated by Karthick S. 8m ago

Details

System Info

Found in Build

Integrated in Build

Property	Value
Browser - Name	Google Chrome 135
Browser - Language	en-IN
Browser - Height	864
Browser - Width	1536
Browser - User agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36
Operating system - Name	Windows NT 10.0; Win64; x64
Operating system - Architecture	x64_64
Operating system - Processor	11th Gen Intel(R) Core(TM) i3-1115G4 @ 3.00GHz
Operating system - Number of processors	4
Memory - Available	814784512
Memory - Capacity	8216240128
Display - Pixels per inch (X axis)	120
Display - Pixels per inch (Y axis)	120
Display - Device pixel ratio	1.25

Discussion

Add a comment. Use # to link a work item, ! to link a pull request, or @ to mention a person.

Project settings

SystemInformation-2025-04-18T03:23:58.168Z.json 1K

8. Test case results

The screenshot shows the Azure DevOps interface for a test plan. The left sidebar contains navigation options like Overview, Boards, Repos, Pipelines, Test Plans, Progress report, Parameters, Configurations, Runs, and Artifacts. The main area displays 'Test Suites' and 'Test Points'. A modal window titled 'TC05 - View Playlist Page' is open, showing 'Test Case Results'.

Outcome	TimeStamp	Configuration	Run by	Tester	Test Plan
Passed	4m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Not Applicable	12m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	14m ago	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Tuesday	Windows 10	Karthikeyan Senthil	Mallu karthick Balaji ...	Music
Passed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Failed	Saturday	Windows 10	Mallu karthick Balaji ...	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music
Passed	Apr 11	Windows 10	Karthick S	Mallu karthick Balaji ...	Music

9. Test report summary

The screenshot shows the Azure DevOps Work Items page for a bug report. The bug is titled 'BUG 203 - BG 01 - Countries Drop down Not Available on the page'. The state is 'New'. The repro steps are listed, and the test results show a 'Failed' status for the last step.

Repro Step

Step no.	Result	Title
1.	Passed	Open Chrome Browser
2.	Passed	Paste the URL (https://127.0.0.1/new.html)
3.	Failed	App should load with new articles

Test Results: 1. Passed, 2. Passed, 3. Failed

- Assigning bug to the developer and changing state

Microsoft Azure Portal | Azure DevOps | My Information | Test Plan 84 Music P | Runs - Test Plans | Settings - Overview

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/runs?_a=resultSummary&runId=48&resultId=100000

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Runs

Run 48 - TS02 - View Playlists (Manual) / TC06 - Playlist Loading Failure

BUG 92 TB01 - Playlist loading spinner keeps spinning indefinitely on poor network

Karthick S 0 comments Add tag

Save & Close Follow Details (3)

Updated by Karthick S 33m ago

Repro Steps

18-04-2025 03:23 Bug filed on "TC06 - Playlist Loading Failure"

Step no.	Result	Title
1.	Failed	Disconnect from internet
		Expected Result
		Network is offline
2.	Failed	Comments: Page Not loading
		Navigate to "My Playlists"
		Expected Result

Error message "Unable to load playlists" is shown

Test Configuration: Windows 10

Planning

Resolved Reason

Story Points

Priority 2

Severity 3 - Medium

Activity

Effort (Hours)

Original Estimate

Remaining

Completed

Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

+ Add link

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Related Work

+ Add link

Add an existing work item as a parent

Tested By

77 TC06 - Playlist Loading Failure

Updated 10-04-2025. # Design

System Info

Comment Page Not loading

Project settings

2. Navigate to "My Playlists"

10. Progress report

Microsoft Azure Portal | Microsoft Azure | My Information | Progress report - Test Plans | KARTHICK-231801079/Music P

dev.azure.com/231801095/Music%20Playlist%20Batch%20Creator/_testManagement/analytics/progressreport

Azure DevOps 231801095 / Music Playlist Batch Creator / Test Plans / Progress report

Music Playlist Batch Cr... +

Overview

Boards

Repos

Pipelines

Test Plans

Test plans

Progress report

Parameters

Configurations

Runs

Artifacts

Project settings

Progress report

Music Playlist Batch Creator - Test Plan

Test Suites

Outcome

Configuration

Tester

Priority

Assigned To

Summary

1 Test plans

14 Test points

14 (14 / 14) Test points run

100% Run

100% (14 / 14) Pass rate

14 Passed

Outcome trend

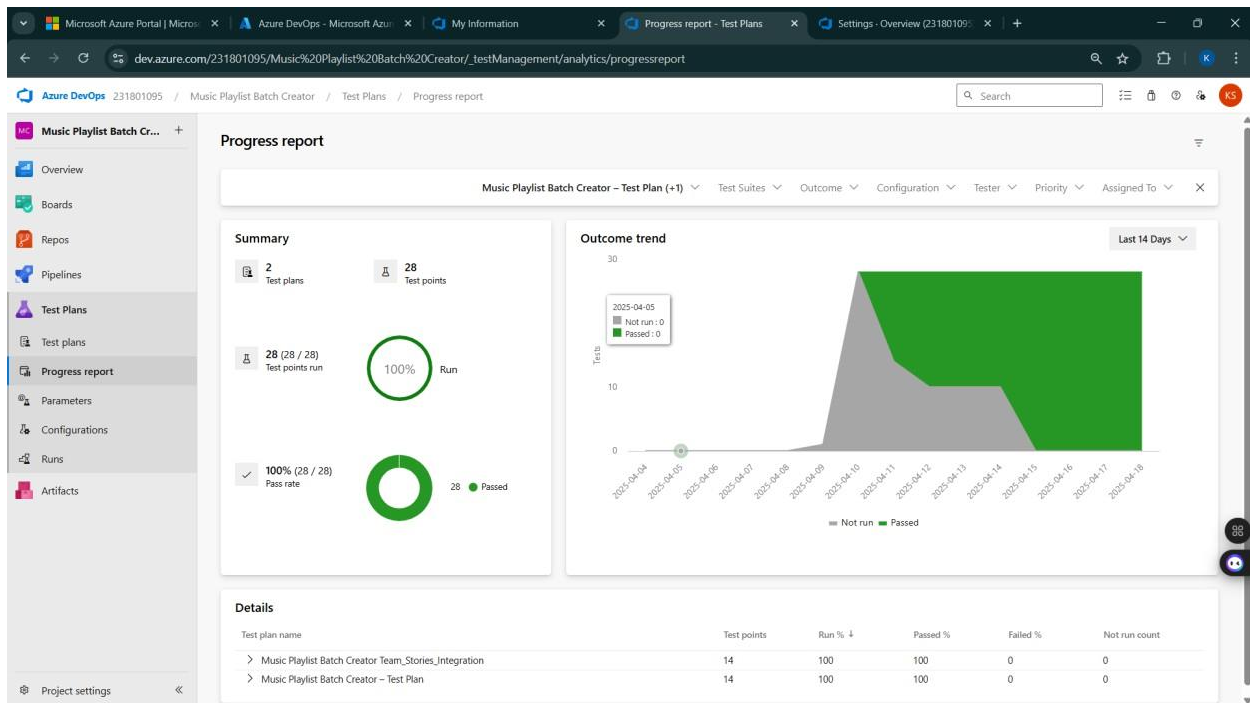
Last 14 Days

Not run

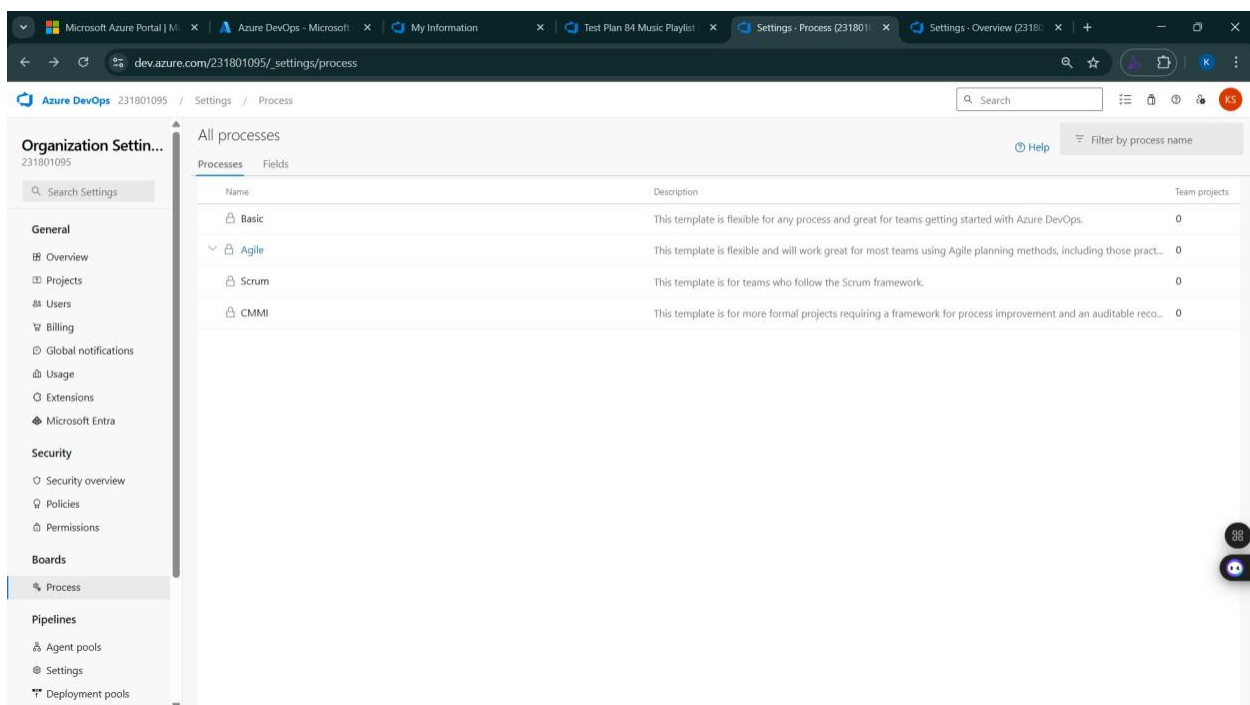
Passed

Details

Test plan name	Test points	Run %	Passed %	Failed %	Not run count
Music Playlist Batch Creator - Test Plan	14	100	100	0	0
TS01 - User Login	4	100	100	0	0
TS02 - View Playlists	2	100	100	0	0
TS03 - Real-Time Metadata	2	100	100	0	0
TS04 - Playlist Editing	4	100	100	0	0
TS05 - Smart Playlist Creation	2	100	100	0	0



11. Changing the test template



Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

All processes

Processes Fields

Filter by process name

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0

Microsoft Azure Portal | M x Azure DevOps - Microsoft x My Information x Test Plan 84 Music Playlist x Settings - Process (231801095) x Settings - Overview (231801095) x

dev.azure.com/231801095/_settings/process

Azure DevOps 231801095 / Settings / Process

Organization Settings 231801095

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools
- Settings
- Deployment pools

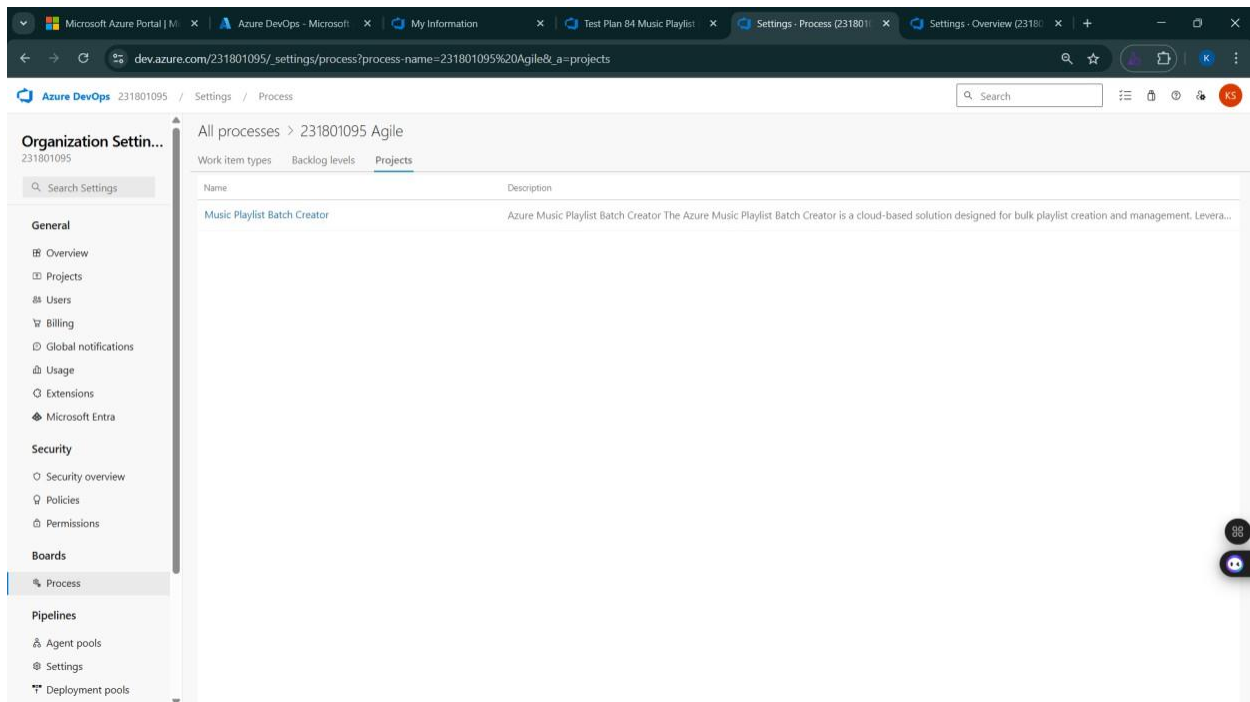
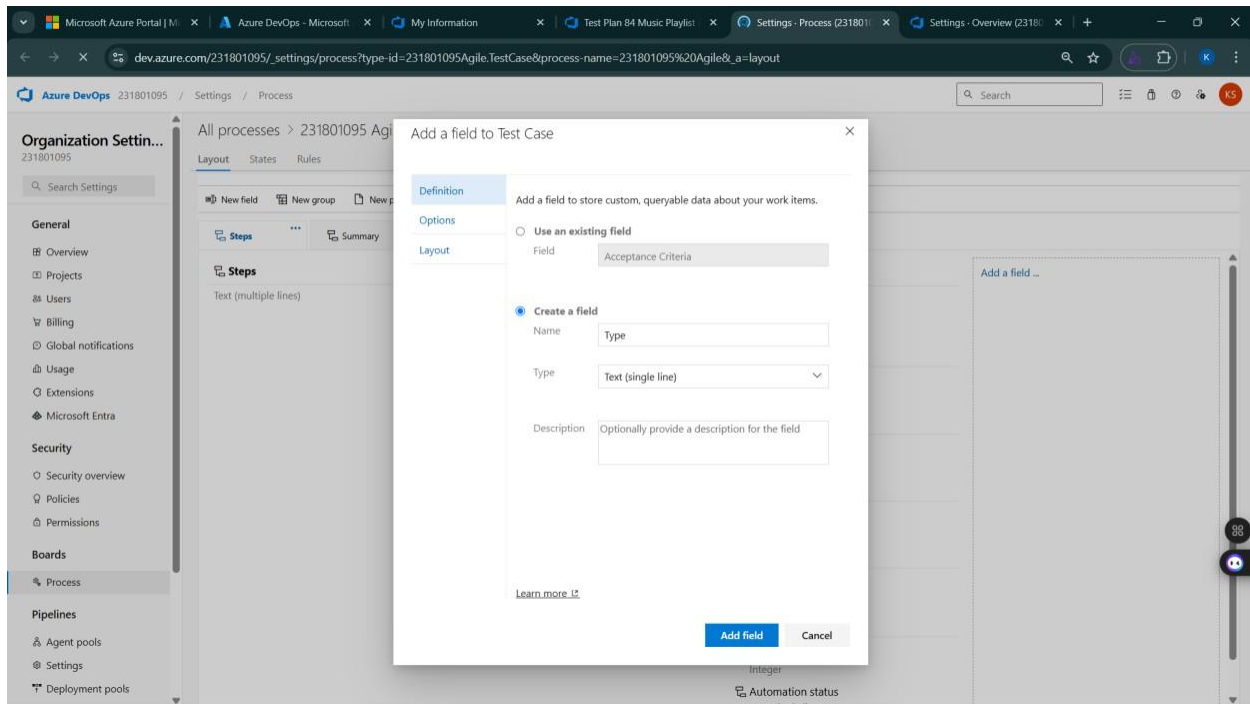
All processes

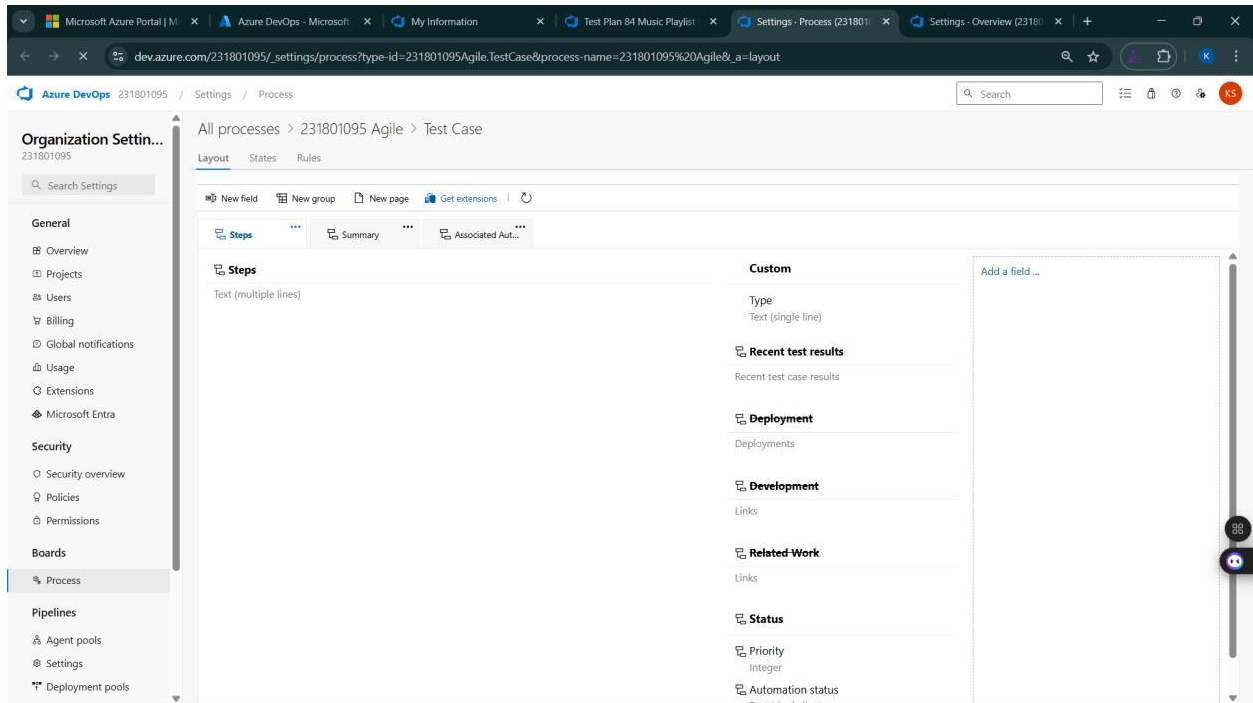
Processes Fields

Filter by process name

Name	Description	Team projects
Basic	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	0
231801095 Agile (default)		1
Agile Plus		0
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record of process.	0

12. View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path